To: Sarah Okafor, Head of E-Commerce
From: Data and ML Team
Subject: “Customers who bought this also bought” and customer groups
Bottom Line: The suggestion box can be refreshed every night in about 2 seconds, from a file of about 4 MB. One important finding: the compressed model on its own gave worse suggestions than simply showing our 10 best sellers. When we blended it with best-seller popularity, it beat the best sellers clearly in our test (35.0% against 27.0%). We recommend trying the blend on a few live product pages first.
What we did and which method we chose: We turned about 684,000 cleaned order lines (up to 8 October 2011) into a table of 5,471 customers by 3,686 products. We picked the settings (50 patterns, blend strength) on an earlier two-month window, then settings chosen on an earlier period, then scored once on the test period on what 1,913 customers bought for the first time in the following two months. The method is called Truncated SVD. It finds 50 “shopping patterns” and describes every customer and product by how strongly it matches each pattern. We chose it because our table is 98% empty, and this method works on such a table directly. We also tried a close relative, PCA: same size, about a second to fit, and slightly more accurate once set up properly (24.6% against 21.8%).
Size and Speed: Stored in full, the table would take 161 MB. Skipping the empty cells brings it to 5 MB, and the compressed version needs 3.7 MB. Most of the saving therefore comes from skipping empty cells, not from the compression itself. The whole nightly job (training, suggestions for every customer, “also bought” lists for every product) takes about 2 seconds (1.3 to 2.3 across runs).
Does it work: The table shows the share of customers for whom at least one of 10 suggestions was something they then bought for the first time. The margin of error is about ±1 percentage point. These scores are for per-customer suggestions; product-page lists use the same model and looked sensible by eye but were not scored.

Method	Success Rate
10 Best Sellers (Baseline)	27.0%
Compressed Model alone	21.8%
Compressed Model + Popularity Blend	35.0%
Compressed alone, same data as row below	23.6%
No Compression (a 109 MB table)	31.9%

What we lose by Compressing: The 50 patterns keep about 38% of the detail in the purchase data: broad buying habits, not one-off choices. Against no compression on the same data, this cost about 8 points (23.6% against 31.9%). The blend more than wins that back (35.0%), but the uncompressed table did not get the same popularity boost, so that comparison is not fair. It also needs 109 MB against 3.7 MB.
Customer Groups:

Group	Share of Customers	Share of Revenue (rounded)
Occasional small buyers	73%	23%
Home-décor and sign shops	20%	34%
Large children’s and kitchen-ware wholesalers	6.5%	31%
Key accounts (12 customers)	0.2%	12.6%

The groups differ mostly in how much they buy and less in what they like. Revenue is for the period up to 8 October 2011, before returns. A chart of the groups is in the slides.
What could go wrong:

1. New customers get no personal suggestions (380 of the 2,422 customers who ordered in the test period were new), and the 23% of order lines with no customer ID could not be used.
2. Products with under 10 buyers are left out (about 1 in 10 test purchases).
3. Suggestions are weaker for niche or all-purpose items.
4. The test period was the Christmas run-up, so results may differ at other times of year.
5. The blend leans towards best sellers, so variety may suffer (not measured).
6. Twelve customers bring 12.6% of revenue, which is a business risk.

Next Steps: Run an A/B test of the blend on a few product pages, give the uncompressed table the same blend to measure the true cost of compressing, and refresh nightly. No customer IDs appear in this memo or in any chart.